

Energy-Efficient MTJ Switching via Electric Field Control

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Voltage-controlled magnetic phenomena have attracted significant interest as a pathway toward reducing the energy consumption of spintronic devices. Among various voltage-controlled mechanisms, voltage-controlled exchange coupling (VCEC) enables direct electrical modulation of magnetic interactions through an applied voltage. Unlike many voltage-induced magnetic effects that rely on ionic motion, VCEC originates from modifications of electronic states and therefore possesses intrinsically fast response characteristics. In this digest, we summarize recent progress on VCEC-based magnetic tunnel junctions (MTJs) and demonstrate how the same physical mechanism can support multiple spintronic functionalities. We first show that VCEC enables continuous tuning of stochastic switching behavior in superparamagnetic MTJs, providing controllable probabilistic outputs suitable for stochastic computing and hardware random number generation. We then summarize nanosecond-scale voltage-pulse-induced switching in perpendicular MTJs, highlighting the rapid response of electron-mediated exchange-coupling modulation. These results establish VCEC as a versatile platform for future low-power memory and computing technologies.

Index Terms—Voltage-controlled exchange coupling (VCEC), Magnetic tunnel junctions (MTJs), ultra fast switching, energy efficient switching, MRAM, Computing

I. INTRODUCTION

Voltage-controlled magnetic phenomena have emerged as promising approaches for reducing the energy consumption of spintronic devices[1], [2]. By manipulating magnetic properties through electric fields rather than large charge currents, voltage-based control can potentially overcome the Joule-heating limitations associated with conventional spin-transfer torque (STT) and spin-orbit torque (SOT) switching.

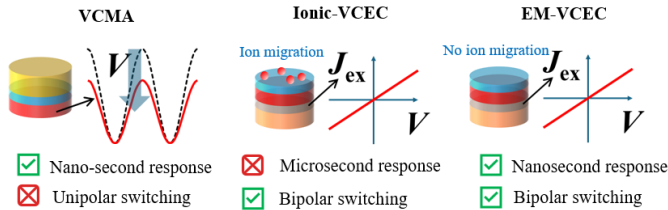


Fig. 1. VCMA offers fast operation but only unipolar modulation. Ionic-VCEC provides bipolar control of exchange coupling via ion migration, at the cost of slow response times. EM-VCEC, achieves bipolar voltage-controlled exchange coupling through electronic modulation without ion migration, retaining the fast nanosecond response while enabling reversible switching.

Several voltage-controlled mechanisms have been investigated in magnetic tunnel junctions (MTJs), including voltage-controlled magnetic anisotropy (VCMA), magneto-ionic effects, charge trapping, and strain-mediated magnetic control. While these approaches have demonstrated efficient manipulation of magnetic states, many involve relatively slow processes such as ionic migration, defect redistribution, or charge relaxation. Voltage-controlled exchange coupling (VCEC) represents a fundamentally different approach. Instead of modifying magnetic anisotropy or relying on atomic motion, VCEC directly alters the exchange interaction between magnetic layers through voltage-induced changes in electronic states[3]. Because exchange coupling is intrinsically electronic

in origin, VCEC offers the potential for both low-energy operation and rapid response.

This digest summarizes these developments and discusses the opportunities offered by VCEC for future low-power memory and computing technologies.

II. PRINCIPLE OF VOLTAGE-CONTROLLED EXCHANGE COUPLING

In a VCEC-MTJ, an applied voltage across the tunnel barrier modifies the exchange interaction between the free layer and the underlying coupling layer, as illustrated in Fig. 1. The resulting variation in exchange energy can be interpreted as a voltage-dependent effective magnetic field acting on the free layer.

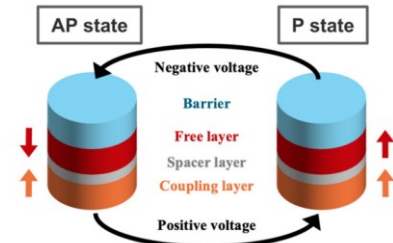


Fig. 2. Schematic illustration of voltage-controlled exchange coupling. An applied voltage modifies the exchange interaction between magnetic layers and produces a voltage-dependent effective magnetic field acting on the free layer.[4]

Unlike conventional spin-transfer torque (STT) switching, which relies on spin-polarized current transfer, VCEC directly modifies the magnetic interaction energy. Consequently, the magnetic energy landscape can be tuned with minimal current flow and reduced Joule heating. Depending on the device configuration and operating regime, exchange-coupling modulation can either alter thermally activated switching probabilities or induce deterministic magnetization reversal.

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A key characteristic of VCEC is that the exchange interaction is electronically mediated. Therefore, the response is governed by changes in electronic states rather than atomic motion. This distinction enables voltage-controlled magnetic manipulation while preserving the intrinsically fast timescale of electronic processes. As discussed in the following sections, this unique feature allows VCEC to support both programmable stochastic operation and high-speed voltage-pulse-induced switching within a common physical framework.

III. VOLTAGE-CONTROLLED STOCHASTIC OPERATION

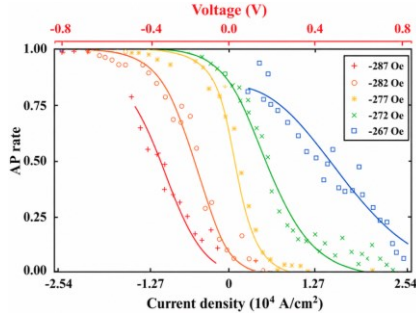


Fig. 3. Continuous tuning of magnetic switching probability through voltage-controlled exchange coupling, demonstrating programmable stochastic behavior in superparamagnetic MTJs.[5]

One important application of VCEC is the electrical control of stochastic magnetic dynamics in superparamagnetic MTJs. When the magnetic energy barrier approaches the thermal energy scale, spontaneous transitions occur between parallel and antiparallel magnetic states. These thermally activated fluctuations generate random telegraph noise (RTN) and form the basis of stochastic spintronic devices. Because VCEC directly modifies the magnetic energy landscape, the occupation probability of each magnetic state can be continuously controlled through applied voltage. Fig. 2 presents experimentally measured switching probabilities under different voltage conditions[5], [6]. As the exchange interaction changes, the probability distribution shifts systematically, demonstrating direct electrical control of stochastic behavior.

Importantly, the observed bipolar shift of the switching probability provides direct experimental evidence of voltage-induced modulation of the effective exchange field. The measured polarity dependence is consistent with theoretical predictions of voltage-controlled exchange coupling. Furthermore, the current density flowing through the MTJ is orders of magnitude smaller than that typically required for spin-transfer torque (STT) switching, indicating that the observed behavior cannot be attributed to STT. In addition, the direction of the effective field change is opposite to that expected from STT-induced switching, further supporting the interpretation that the probability modulation originates from voltage-controlled exchange coupling.

IV. FAST ELECTRONIC RESPONSE OF VOLTAGE-CONTROLLED EXCHANGE COUPLING

Beyond stochastic operation, the electronic nature of VCEC suggests the possibility of rapid magnetic response under voltage excitation. To investigate this behavior, voltage-pulse-induced switching experiments were performed on

perpendicular MTJ structures incorporating exchange-coupled magnetic layers. Fig. 3 shows the measured switching probability as a function of pulse width under different magnetic field conditions. Reliable unipolar switching is observed using nanosecond-scale voltage pulses.

The significance of this result extends beyond the switching speed itself. The observed behavior demonstrates that exchange-coupling modulation remains effective under short-pulse operation and confirms that the underlying magnetic interaction responds on timescales relevant for high-speed device applications.

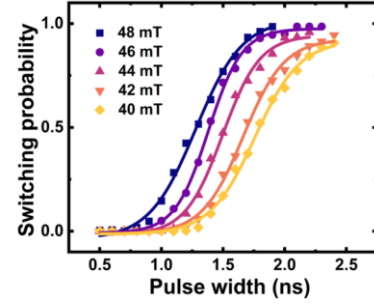


Fig. 4. Voltage-pulse-induced unipolar switching in a perpendicular MTJ. The observed nanosecond-scale response highlights the fast electronic nature of voltage-controlled exchange coupling.[4]

Unlike magneto-ionic switching mechanisms, which require physical motion of ions or oxygen vacancies, VCEC originates from modifications of electronic states associated with exchange coupling. Consequently, its ultimate response time is not fundamentally limited by slow atomic processes. This distinction is particularly important for future memory technologies that require both low energy consumption and fast operation[7]. The demonstrated nanosecond-scale switching provides experimental evidence that VCEC is compatible with high-speed magnetic manipulation while retaining the advantages of voltage-based control. Further optimization of coupling structures and device geometry may enable even faster operation in future generations of VCEC devices.

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